

Codex Relatio: A Descriptive Framework for Analyzing Relational Dynamics in Human–AI Interaction

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Abstract

AI systems increasingly participate in long-term relational contexts such as coaching, companionship, and care. Yet most Human–AI Interaction and AI ethics frameworks focus on interaction-level behaviors or normative evaluation, leaving limited vocabulary for describing relational dynamics as structured systems prior to ethical judgment. We introduce Codex Relatio, a descriptive, non-normative framework for mapping relational configurations across five interdependent axes: Persuasion, Consent, Intimacy, Affection, and Responsibility (PCIAR). The framework models relationships as multiaxial, temporally evolving configurations whose dynamics exhibit feedback loops and cascading effects. By providing a pre-normative descriptive vocabulary, Codex Relatio supports systematic analysis of relational positioning and emerging risks in asymmetric Human–AI contexts without prescribing ideal relational states.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Walkthrough evaluations; Interaction design; Interaction design theory, concepts and paradigms; • **Computing methodologies** → Artificial intelligence; Philosophical/theoretical foundations of artificial intelligence; Theory of mind.

Keywords

relational dynamics, Human–AI interaction, AI alignment, descriptive framework, AI ethics, relational risk detection, relational competence

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1 Introduction

Although relationships are central to social organization, they are not always formalized as structured systems for analysis in the existing literature. Even in HCI, where context and situated action are emphasized [22.], relational structures remain largely implicit,

creating a gap in HAI. Although critical HCI highlights normative aspects of interaction [3.], it frequently lacks the descriptive tools necessary to disentangle relational structure from moral evaluation.

By framing power as a dynamic relational configuration [9.], we can observe how AI systems operate under asymmetry, opacity, and scale, frequently affecting users and collectives over time. Prior work has shown how users negotiate expectations and authority in ongoing interactions with systems [2.]. Without explicit representations of these dynamics, systems lack mechanisms to reason about how their actions propagate through relational systems, leading to unintended effects such as over-dependence, consent erosion, or displacement of responsibility.

This gap is particularly relevant to AI alignment, where linking local actions to broader relational consequences remains an ongoing challenge. Abstraction gaps in sociotechnical systems often obscure downstream relational consequences [19.], particularly when relational dynamics are treated as homogeneous or universal. In this context, existing HAI guidelines focus primarily on interaction-level behaviors rather than relational dynamics unfolding over time [1.]. In the absence of a shared descriptive framework, treatments of relational dynamics may become standardized in ways that overlook contextual variation. Consequently, these systems struggle to recognize or analyze contextual peculiarities and inadvertently propagate biases.

As AI systems gain increasing cognitive capacity without corresponding relational modeling, socio-technical alignment issues may intensify. This work does not argue that AI systems should emulate human relationships, but that they must reason about their relational positioning and analyze other relationships without normative assumptions. Without this, systems cannot contextualize their actions socially, limiting user experience and integration.

This work makes the following contributions:

- Introduces a descriptive, non-normative relational framework for HCI and HAI.
- Formalizes a multidimensional model enabling analysis of relational dynamics across complex contexts.
- Explores implications for relational AI design, addressing unintended consequences before deployment.

2 Background and Conceptual Gap

2.1 Relational Approaches in HCI

HCI research has long examined interaction as situated practice [7., 22.], yet relational structures are rarely formalized as systems within mainstream models. Practice-based approaches foreground activity and context [12.] but provide limited means to account for how local interactions propagate through broader relational



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structures over time. This limitation is critical in HAI contexts with recurrence and long-term engagement, where elements like power, dependency, trust, and agency operate systemically.

2.2 AI Ethics and Normative Shortcuts

AI ethics has primarily focused on universal normative principles [8.]. While valuable, these approaches often compress heterogeneous relational configurations into generalized moral categories. This tendency mirrors, in part, jurisprudence, where fact-finding precedes judgment; however, in HAI, no systematic descriptive phase captures relational structure before evaluation. Without such grounding, ethical reasoning risks abstraction from situated sociotechnical dynamics, particularly in asymmetric contexts [14.].

2.3 Relational Reduction and Systemic Gap

Across ethics, affective modeling, and interaction design, existing approaches provide rich localized accounts yet lack a shared descriptive vocabulary for relational configurations as evolving systems; affective computing similarly reduces relational persistence to momentary signals [4., 15.]. HAI guidelines likewise emphasize interaction-level behaviors [1.], leaving relational topology—roles, obligations, dependency, and power—structurally unarticulated. Consequently, AI systems remain limited in analyzing relational consequences prior to normative evaluation or intervention. Addressing this limitation requires a descriptive framework independent of moral judgment— an objective this work aims to advance.

3 Approach: From Tacit Relational Knowledge to Systemic Modeling

3.1 Hospitality as a Relational Domain

Hospitality is structurally defined by an irreducible tension between unconditional openness and the necessity of limits, forming a foundational basis for human relations [6., 16.]. Its heuristic value lies in operating as a meta-practice across asymmetric relational contexts — institutionalized and historically codified for managing high-stakes encounters under conditions of uncertainty [10., 18.]. By formalizing roles, boundaries, and obligations, hospitality makes tacit relational dynamics observable through structured, negotiated responses that mediate trust, safety, and long-term relational stability. Compared with broader relational theories, which often remain abstract, hospitality exposes latent relational tensions in ways that are empirically accessible, providing a tractable analytical domain for examining systemic relational dynamics across diverse contexts.

3.2 Tacit Relational Knowledge

Relational understanding is often embedded in practice rather than formalized in explicit analytical models [11., 17.]. Practices such as those observed in hospitality provide a lens into tacit relational reasoning, which emphasizes continuity, role-based responsibility, and the systemic consequences of local actions over time [13.]. The recurring mechanics observed across these practices — boundary negotiation, asymmetric consent, and affective calibration — provided the empirical basis from which the PCIAR axes were abstracted. While HCI has formalized interactional and behavioral models, it has engaged less directly with structured accounts of

relational reasoning. This gap suggests an opportunity to translate practice-based insight into analytical form, capturing relational dynamics without reproducing culturally specific norms.

3.3 Analytical Translation and Functionalist Stance

Recurring relational mechanics identified across domains were abstracted into a systemic framework through analytical translation of tacit relational reasoning [20., 21.]. The development process was guided by three criteria:

- Coherence, integrating multiple relational dimensions into a unified structure;
- Generality, enabling cross-context application without reliance on culturally specific prescriptions;
- Non-normativity, maintaining separation between descriptive mapping and ethical evaluation.

These criteria shape the framework as a pre-normative diagnostic scaffold, meant to support careful observation and analysis rather than serve as an empirical closure. From a functionalist perspective, the relational axes are treated as observable and operational dimensions, focusing on descriptive clarity and diagnostic value, while remaining agnostic regarding ontological claims about agency or sentience [5.].

4 Codex Relatio: A Multidimensional Descriptive Framework

4.1 Overview of Codex Relatio

Codex Relatio provides a descriptive, non-normative framework for mapping relational structures across Human–Human and Human–AI interactions. Analogous to the “instruction phase” in jurisprudential fact-finding, it identifies the essential relational dimensions and provides a vocabulary to analyze relational configurations and dynamics across agents, without prescribing normative outcomes.

4.2 Relational Typology and Axes

Relational typology refers to the network of connections between agents that shape patterns of social integration and participation. Relationships are organized into systemic orders based on degrees of individual participation, rather than predefined categories. Each relational bond comprises five interdependent dynamic axes (PCIAR):

- *Persuasion*: Degree of cognitive influence, from neutral information to coercive pressure.
- *Consent*: Strength of boundaries and ability to negotiate relational terms.
- *Intimacy*: Depth of access to private domains (physical, emotional, intellectual).
- *Affection*: Observable relational behavior, independent of ontological source.
- *Responsibility*: Accountability across agents (legal/social, situational, elective).

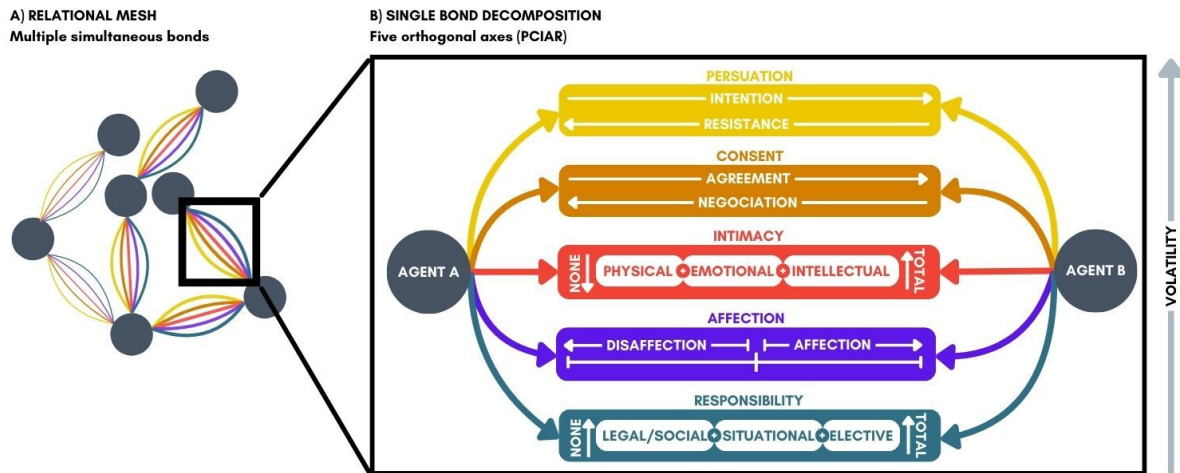


Figure 1: Relational Mesh and Axes Decomposition in Codex Relatio. Left: Relational mesh showing multiple simultaneous bonds (line thickness = bond force; colors = PCIAR axes). Right: Single bond decomposed into five bidirectional dimensions with sub-components.

4.3 Systemic Considerations and Relational Dynamics

By organizing relational dynamics through PCIAR axes, *Codex Relatio* makes otherwise latent behavioral patterns observable and analyzable over time. This enables more accurate and nuanced assessment of how behaviors propagate through relational systems, without prescribing normative judgments. The systemic perspective supports examination of multiaxial interactions, asymmetries, and temporal shifts, particularly in Human–AI contexts where relational configurations evolve without clear social scripts.

5 Discussion and Implications

5.1 Implications for HCI Design

Codex Relatio enables visualization of contrasts between normative concepts and observed relational behaviors, making previously implicit dynamics empirically accessible. By monitoring axis convergence (e.g., high Persuasion with collapsing Consent), systems could anticipate relational misalignments.

Illustration 2 depicts this detection mechanism using a radar chart comparing the alignment model reference and an illustrative sycophancy occurrence in a HAI context. Visualization reveals multiaxial abnormalities: Persuasion, affection and Intimacy exceed reference thresholds while Consent and Responsibility collapse below safe levels. This pattern—elevated influence paired with weakened boundaries—constitutes a systemic alarm signaling the need for intervention before relational harm occurs.

5.2 Implications for HAI

While traditional UX constructs such as usability, usefulness, and affective experience capture interaction-level qualities, Codex Relatio operates at a systemic relational layer, enabling analysis of how these qualities accumulate, drift, or destabilize over time across asymmetric configurations.

Consider a user engaging with an AI mental health companion over several weeks. Initially, Intimacy remains low as the user shares surface-level concerns, while Consent is high and Responsibility is distributed. As the system adapts to the user’s patterns, Intimacy and Persuasion gradually increase as the system reinforces certain interaction patterns. Consent is not explicitly renegotiated as relational depth shifts, and perceived responsibility may migrate toward the system.

This multiaxial drift—rising Intimacy and Persuasion alongside static or eroding Consent—could be identified as a potential systemic alarm through PCIAR analysis before relational harm occurs. By decoupling relational competence from biological sentience, Codex Relatio supports analyzing AI systems as relational actors without anthropomorphic assumptions. The framework also provides a vocabulary for describing asymmetries of responsibility, highlighting relational risks such as manipulative coaching or AI gaslighting.

Although the framework does not prescribe interventions, the descriptive mapping of relational configurations could inform design exploration, helping researchers and practitioners consider boundaries, alignment, and relational risk across diverse HAI contexts.

5.3 Implications for Research

Codex Relatio provides a descriptive framework that could inform future research on social and relational computing. Its orthogonal axes offer a basis for exploring relational metrics (e.g., consent-to-persuasion ratios), which could support comparative studies of LLMs and other social agents. The framework also offers a structured lens for analyzing large-scale interaction logs, allowing researchers to investigate how different relational topographies, historical or situational, might relate to trust, stability, and alignment over time. *Codex Relatio* does not replace existing interaction or ethical frameworks but could complement them by providing a

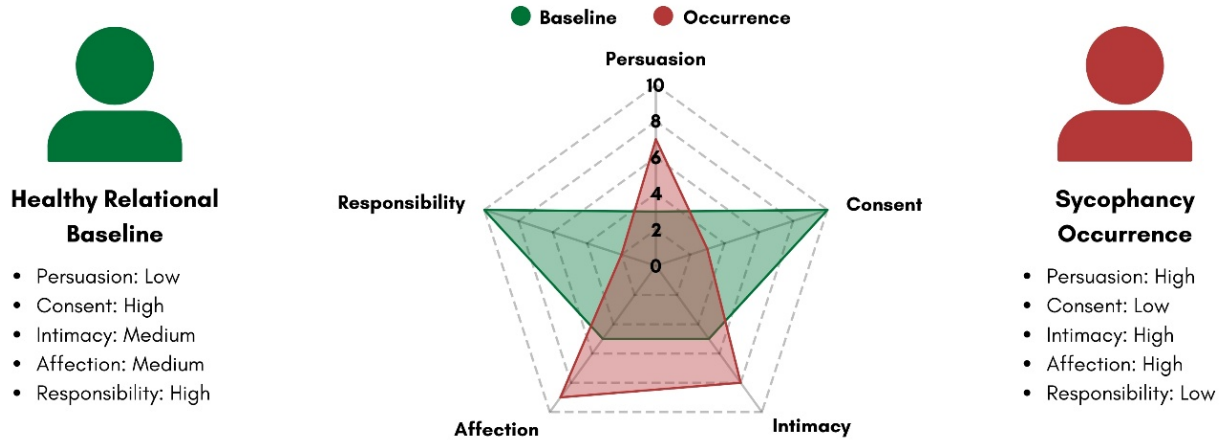


Figure 2: Relational Risk Detection via Radar Chart Analysis. Comparison between Healthy Relational Baseline (green) and Sycophancy occurrence (red) in a Human-AI relationship. Multiaxial abnormalities—high Persuasion and Intimacy with collapsed Consent and Responsibility.

pre-normative vocabulary for describing relational structure prior to normative evaluation or design intervention.

6 Limitations and Future Work

The current iteration of *Codex Relatio* is primarily derived from qualitative analysis and the synthesis of hospitality-based tacit knowledge. While the structural axes are robust, the framework has not yet been empirically tested across large-scale or diverse demographic datasets. Future work could explore validating relational types and their descriptive integrity across cultural and linguistic contexts. Potential directions include observational studies, exploratory computational modeling of PCIAR axes, or cross-cultural analyses of hospitality-inspired relational calibration.

7 Conclusion

This work introduces *Codex Relatio* as a descriptive, pre-normative framework for analyzing relational dynamics in HAI. By providing a structured vocabulary for relational structure, the framework supports reflection on relational positioning prior to ethical evaluation. Future work could explore longitudinal studies of how relational axes evolve over time, as well as applications for connecting descriptive insights to design considerations. These directions build on the framework's conceptual foundation, emphasizing observation and analysis rather than immediate operational deployment.

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